KrishiGrid Bangladesh
Data-Driven Food Security Simulation

KRISHIGRID BANGLADESH

A Data-Driven Simulation of Food Import Dependency and Decentralized Agricultural Infrastructure

PART 4 — THE SOLUTION MODEL

How Bangladesh can convert agricultural potential into real production, income, and stability ?

The issue is not how much we produce , It is how poorly production connects to markets

Introducing "KrishiGrid" a ward level decentralized agricultural infrastructure network

Analysis : System design, infrastructure gaps, cost optimization
Tools : SQL, Power BI,Canva
Datasets : Integrated insights from Parts 1 to 3

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THE SOLUTION MODEL

WHY DO WE NEED KRISHIGRID?

FROM PART 1 — THE IMPORT CRISIS

Bangladesh spent over $70 billion importing food.
Edible oil dependency remains around 90 percent.
Currency depreciation increased real import cost significantly.
Even a 25 percent reduction could save over 12,000 crore BDT annually.

FROM PART 2 — THE HIDDEN LAND

Over 1.69 million hectares of land remain underutilized.
Rice fallow land alone exceeds 800 thousand hectares.
Cropping intensity is already near 200 percent in many regions,but efficiency gaps still exist.
Untapped land is highly concentrate.

FROM PART 3 — THE CROP REVOLUTION

Bangladesh can generate ~12.7 million MT of additional crops.
Edible oil potential reaches ~2.0 million MT.
Mustard Rapeseed alone contributes over 1.08 million MT.
Short duration crops can unlock additional crop cycles.

The land exists.
The crops exist.
The farmers exist.

THE MISSING PIECE

What is missing is localized infrastructure that connects production to processing, storage, and markets at the village level.

KRISHIGRID IS THAT MISSING SYSTEM

QUESTION: What is KrishiGrid Ward Hub?

The KrishiGrid Ward Hub is an integrated rural facility that brings storage, processing, market access, and farmer support into one location within walking distance. It includes warehouse and small cold storage to reduce post harvest losses, along with a seed and fertilizer bank ensuring timely access to quality inputs.
The hub enables value addition through oil, rice, and flour mills, along with drying facilities, allowing farmers to earn more from their produce. The collection and market zone acts as a local agricultural marketplace, combining collection, sorting, and direct farmer to buyer trade in one space to improve price realization.
It also provides training and advisory support through a training center and help desk. Each hub serves around 1500 to 2500 farmers, operates with 7 to 10 staff, and is located within 1 kilometer for easy access.

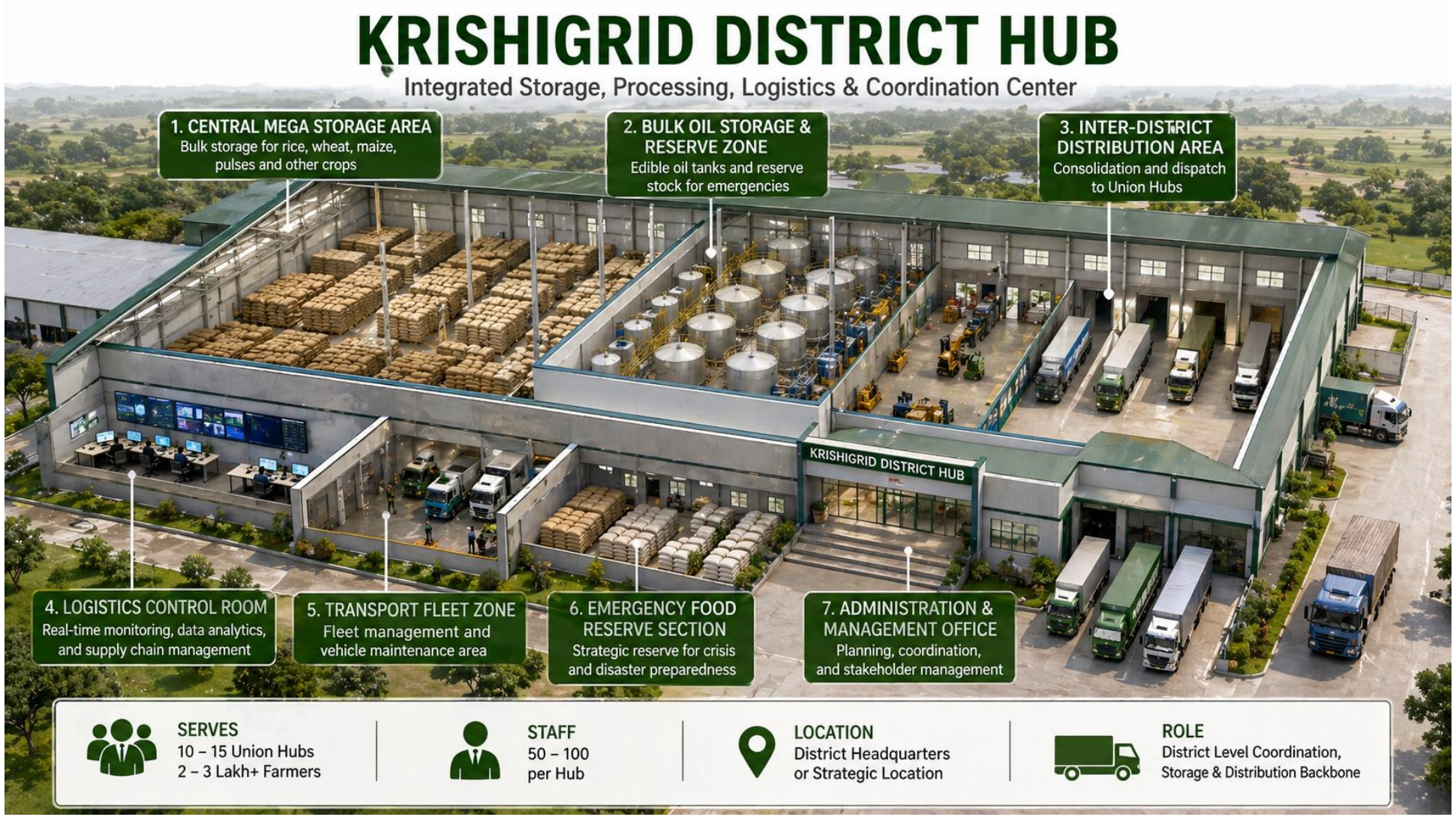
QUESTION: What is KrishiGrid Union Hub?

The KrishiGrid Union Hub acts as the coordination and distribution layer of the system, connecting multiple ward hubs into a unified network. It focuses on large scale storage and efficient movement of goods rather than local level processing or retail activities.

The hub includes bulk storage facilities for grains, oilseeds, and other crops, along with structured loading and unloading zones to support smooth logistics. It manages aggregation of produce from ward hubs and ensures organized distribution across regions.

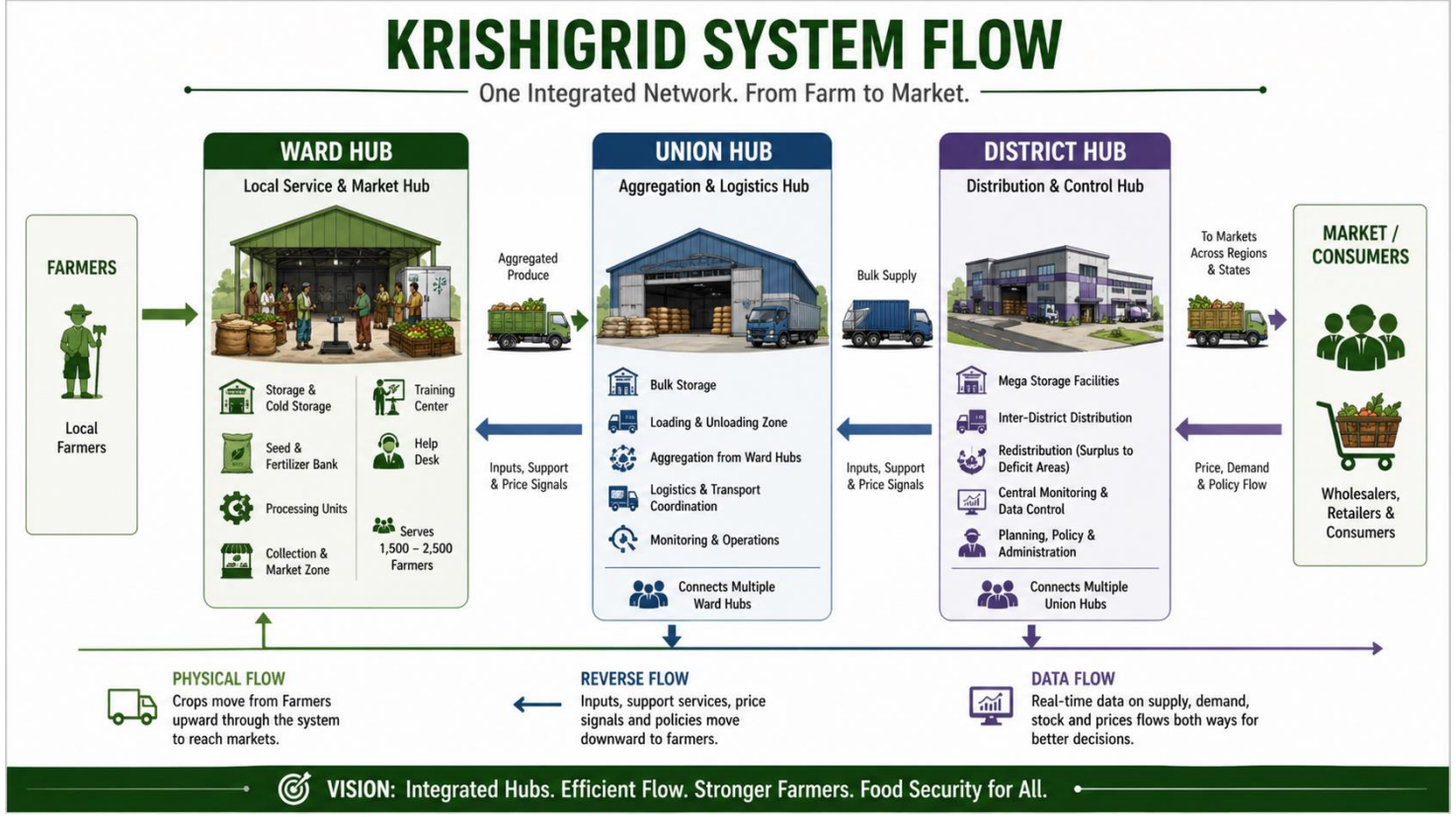
A central monitoring and control system tracks supply flow, stock levels, and movement efficiency, while administrative functions coordinate operations between wards and higher level hubs. This layer strengthens the supply chain by reducing fragmentation and improving consistency.

Each union hub connects multiple ward hubs, acting as the logistical backbone that enables scalable and efficient agricultural distribution.



QUESTION: What is KrishiGrid District Hub?

The District Hub functions as the strategic control layer of the system, managing large scale storage and overseeing inter regional movement of agricultural goods. It connects multiple union hubs and ensures balanced distribution from surplus to deficit areas through coordinated logistics. With centralized monitoring of stock, demand, and supply flow, it enables faster decision making and reduces inefficiencies across the network. This layer strengthens overall system stability and plays a critical role in ensuring long term food security at the national level.



KRISHIGRID SYSTEM FLOW

The KrishiGrid system operates through three interconnected flows that move products, information, and value across the entire agricultural network.

1. Physical Flow (Upward Movement)

Crops move from farmers to Ward Hubs, where they are collected, processed, and partially stored. From there, produce is aggregated at Union Hubs and then transported in bulk to District Hubs for large-scale storage and redistribution. Finally, goods are supplied to markets, wholesalers, and consumers.

👉 This ensures efficient movement from farm to market with minimal loss and better logistics coordination.

2. Reverse Flow (Downward Support System)

Inputs, services, and decision signals flow in the opposite direction. District and Union Hubs send price signals, demand insights, policies, and logistics support down to Ward Hubs, which directly guide farmers.

👉 This transforms farmers from isolated producers into informed decision-makers.

3. Data Flow (Two-Way Intelligence Loop)

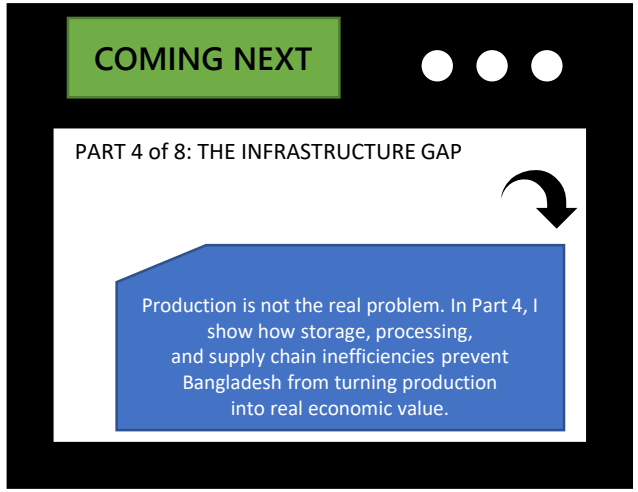
Real-time data on production, demand, prices, and stock levels flows both upward and downward across all hubs. This continuous feedback loop enables better planning, prevents shortages, and balances supply across regions.

👉 This is what makes the system “smart,” not just physical infrastructure.

KEY DESIGN PRINCIPLES

Not every hub requires every machine
Each hub is built based on regional crop needs
This reduces unnecessary investment and improves efficiency.

- Ward level access instead of union level 1 km distance enables ~90 percent participation vs ~30 percent.
- Rural hubs focus on production and processing Cities operate as sales and distribution outlets.
- Inter hub food grid balances surplus and deficit across districts.
- Government backed hub pricing reduces dependence on middlemen.



WHY THIS DESIGN WORKS

1 PROXIMITY

Farmers operate within 1 km access
Participation rises to ~90 percent compared to distant systems

2 CUSTOMIZATION

Different regions grow different crops
Avoiding one size fits all prevents major resource waste

3 PROCESSING AT SOURCE

Farmers process crops locally instead of selling raw output
Value addition directly increases farmer income

4 PRICE ANCHOR

Local hub pricing creates a fair benchmark
Market prices adjust when farmers have a direct alternative

5 FOOD GRID

Surplus flows between regions dynamically
Eliminates artificial shortages and supply imbalance